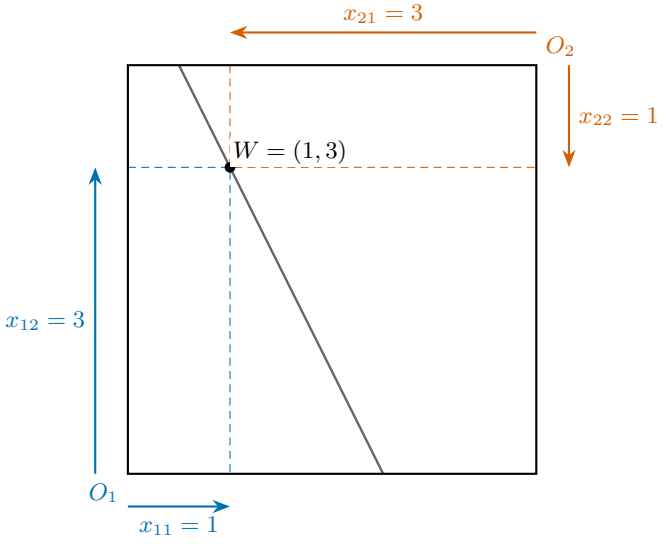

From O_1: $x_1 = (1, 3)$

From O_2: $x_2 = (3, 1)$

Totals remain $(4, 4)$

$p = (2, 1)$
Shared boundary: $2a + b = 5$
Slope: $-p_1/p_2 = -2$

Consumer 1: $2a + b \leq 5$
Consumer 2: $2a + b \geq 5$